

Large-Scale and Multi-Structured Databases

Introduction to the Course

Academic Year 2027-2027

Prof Pietro Ducange

Who is Talking to You?

Pietro Ducange

- Born in Apulia, South of Italy
- Master Degree in Computer Engineering in 2005, University of Pisa
- PhD in Information Engineering in 2009, University of Pisa
- Post-doc Researcher 2009-2014, University of Pisa
- Associate Professor 2014-2019, eCampus University
- Associate Professor 2019-on going, University of Pisa

Pietro's Research Activity

Main Research Topic:

- Big Data Mining and Analytics
- Text Analysis
- Explainable Artificial Intelligence

Member of:

AI&RD Research Group @ DDI

<https://ai.dii.unipi.it>

Cloud Computing, Big Data and Cyber Security Lab@DII:

<https://crosslab.dii.unipi.it/cloud-computing-big-data-cybersecurity-lab>

Publication Records:

<https://scholar.google.it/citations?user=HCgZqXEAAAAJ&hl=it>

The Course

Large Scale and Multi-Structured Databases

9 CFU-> 90 Hours

Program Degrees:

- M.Sc. in Artificial Intelligence and Data Engineering (1-2 Year)
- M.Sc. in Computer Engineering (1 Year)

Timetable for the next to week

Schedule Changes

Large-Scale and Multi-Structured Databases



Some class slots are changed in coordination with the course
Performance Evaluation of Computer Systems and Networks.

Date	Time	Originally scheduled for Large-Scale	Change	New schedule for Large-Scale
Tue 22 Sep	14:30 – 16:30	Large-Scale	Cancelled	No class
Wed 23 Sep	11:30 – 13:30	Large-Scale	Swapped with Performance Evaluation	Performance Evaluation (instead of Large-Scale)
Wed 23 Sep	15:30 – 18:30	Performance Evaluation	Swapped with Large-Scale	Large-Scale (instead of Performance Evaluation)
Tue 29 Sep	11:30 – 13:30	Large-Scale	Replaced by Performance Evaluation	Performance Evaluation (instead of Large-Scale)
Tue 29 Sep	14:30 – 16:30	Large-Scale	Cancelled	No class
Wed 7 Oct	15:30 – 18:30	Performance Evaluation	Assigned to Large-Scale	Large-Scale (instead of Performance Evaluation)
All other dates	—	—	No changes	As in the original timetable

- These changes **only affect** the classes of **Large-Scale and Multi-Structured Databases** and **Performance Evaluation of Computer Systems and Networks.**
- All other classes remain as scheduled.

See you in class!

Syllabus

Introduction and Motivations: Introduction to the Course, The Big Data Era, The Database Revolutions

Fundamentals and properties of the NoSQL databases: ACID vs BASE properties, The Cap Theorem, Scalability, Sharding, Replication, Consistency

Architectures of NOSQL databases: Document Databases, Key-values Databases, Column Databases, Graph Databases

Recaps: Recap of Software Engineering and Java (basics, connections and queries towards SQL databases).

Modern Infrastructures for NoSQL Databases: REDIS, MongoDB, Neo4J (Installation, configuration, CRUD operations, main queries)

The Data Base Revolutions

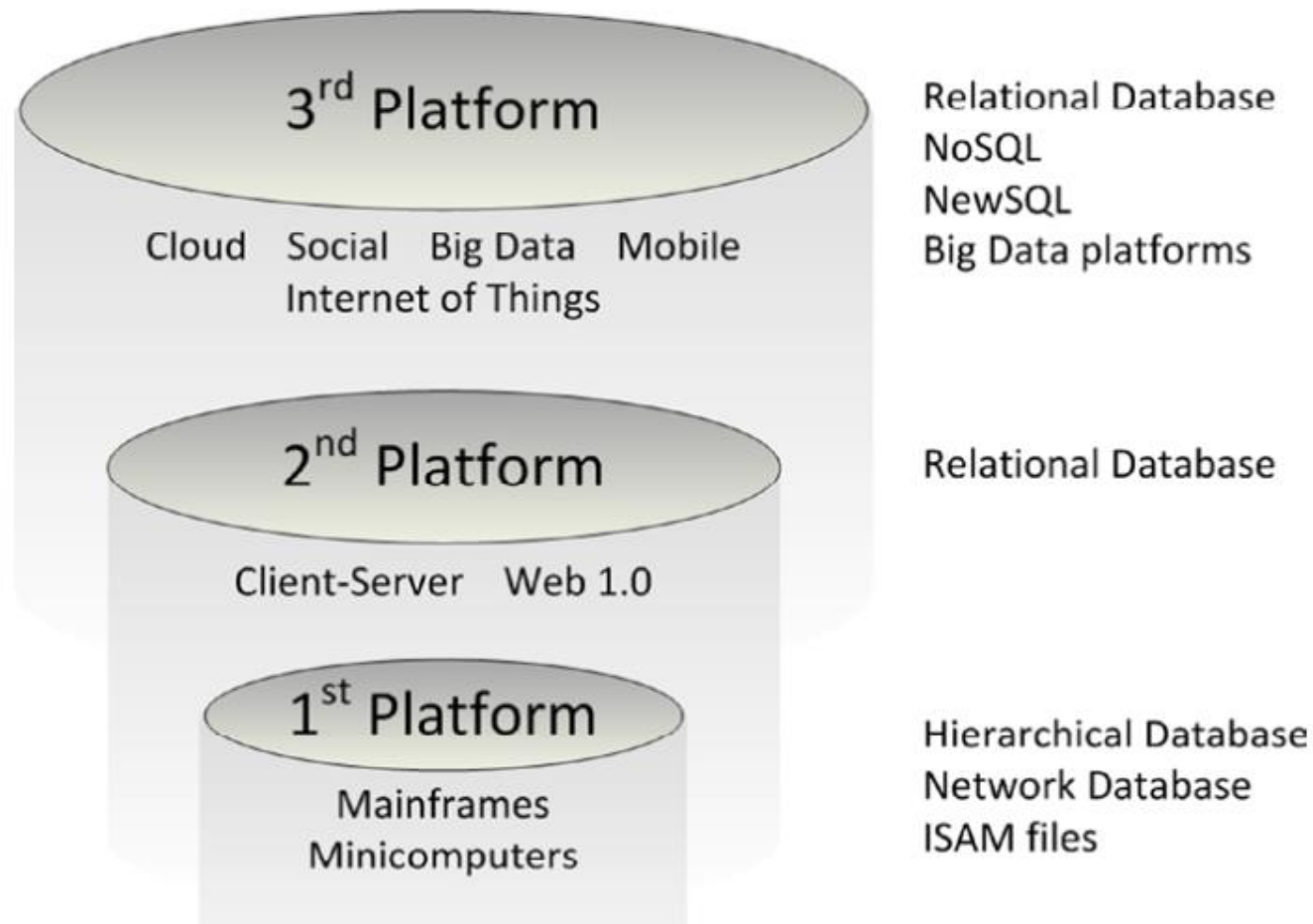


Image extracted from "Guy Harrison, Next Generation Databases, Apress, 2015"

The Big Data Era

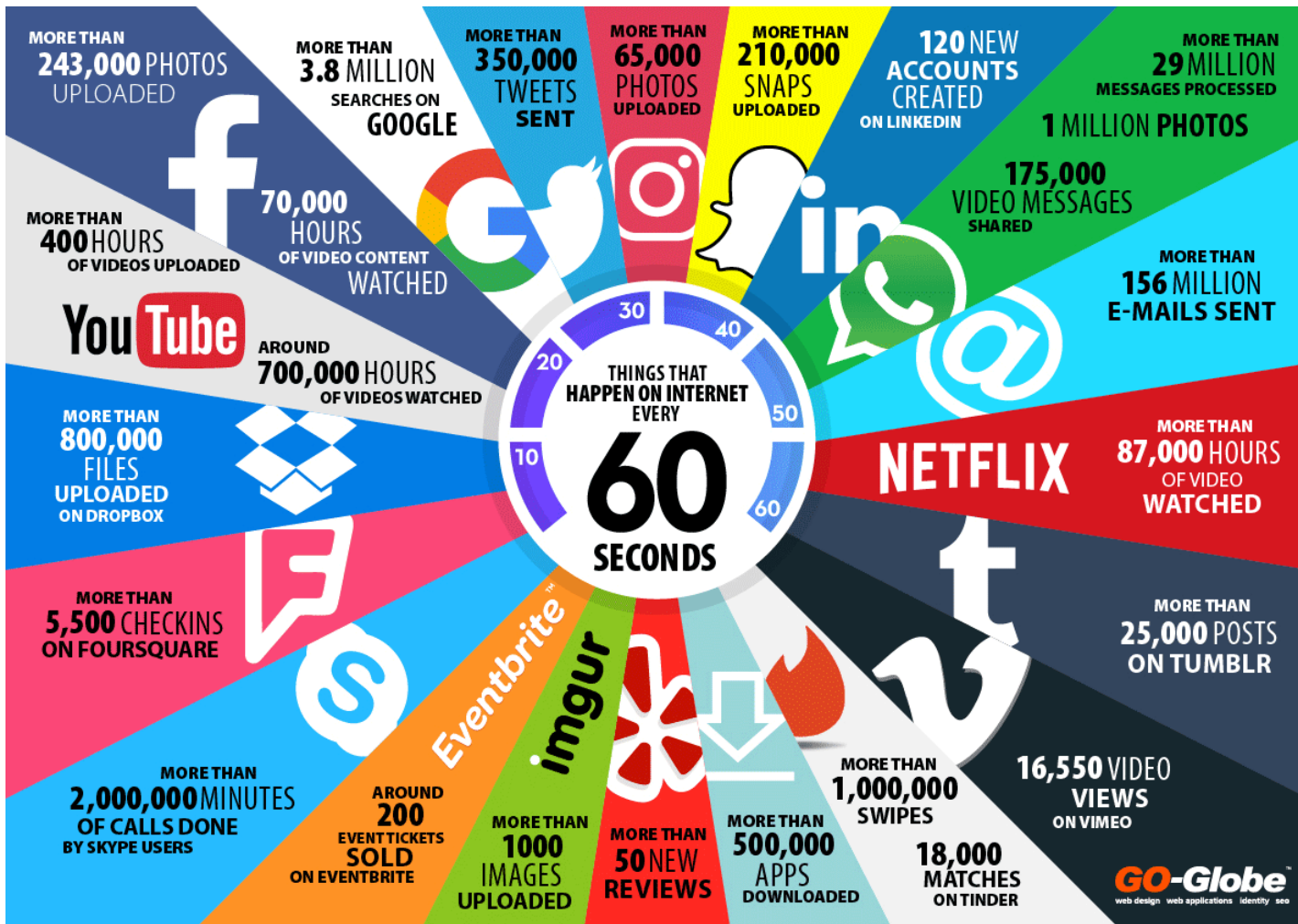


Image Extracted from: <https://www.go-globe.com/things-that-happen-on-internet-every-60-seconds/>

ACID vs BASE

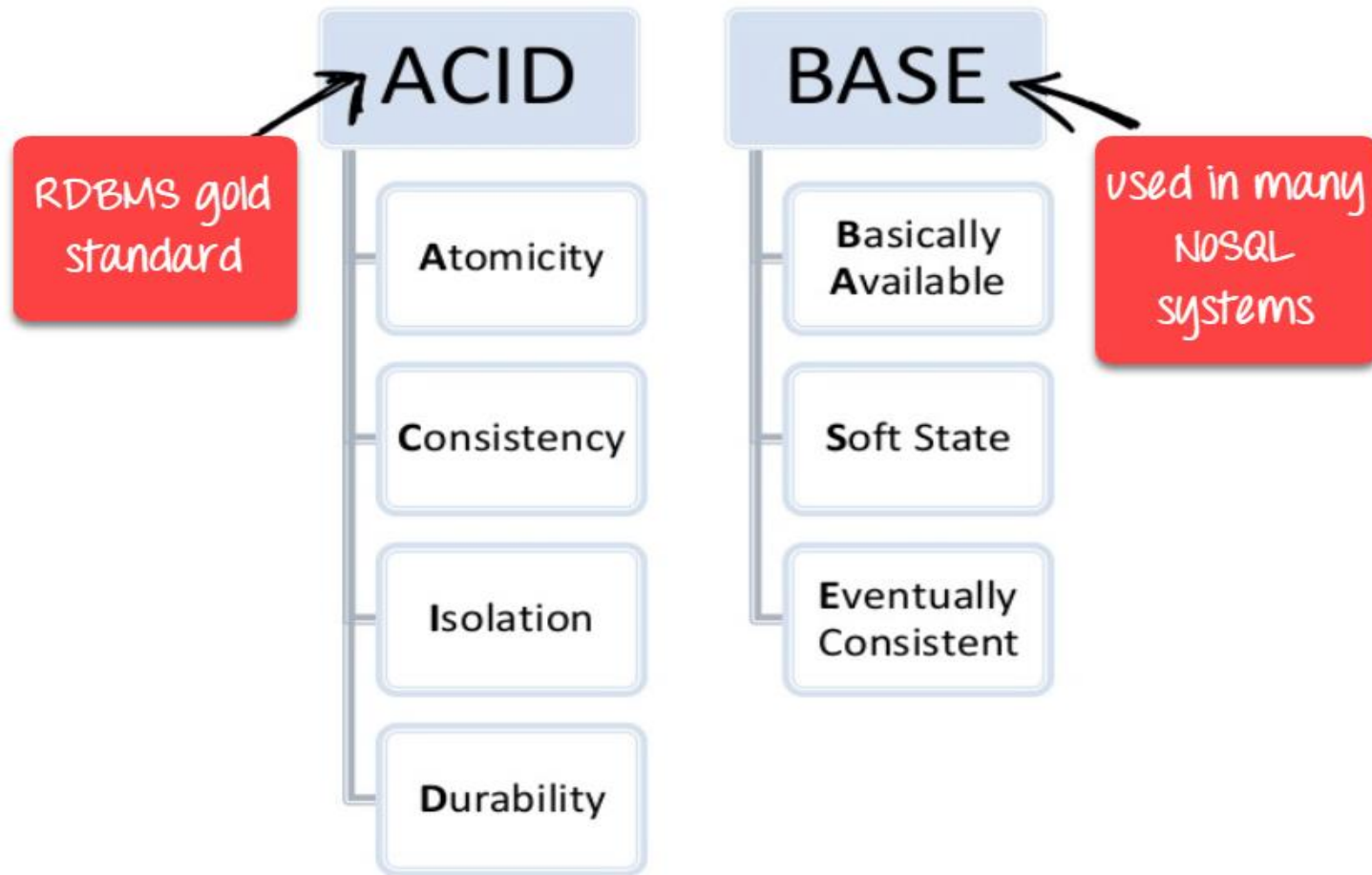


Image extracted from: <https://www.guru99.com/sql-vs-nosql.html>

Key-Value Databases

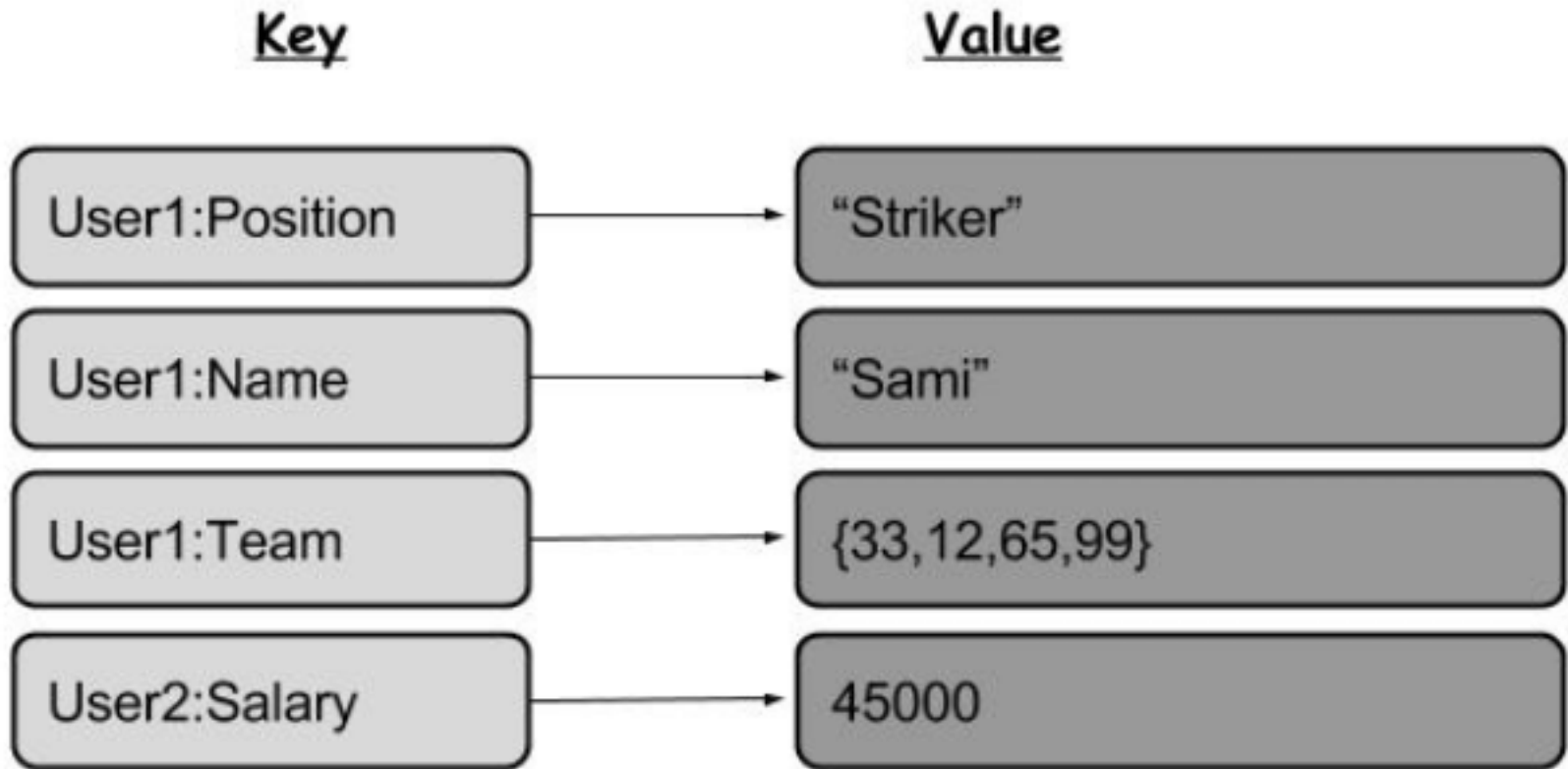
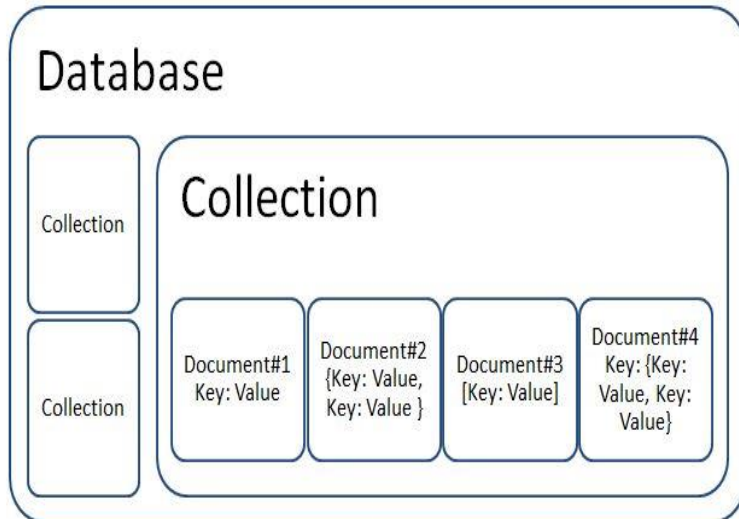


Image extracted from: https://www.researchgate.net/figure/Key-value-NoSQL-Database_fig1_332188615

Document Databases



Document 1

```
{
  "id": "1",
  "name": "John Smith",
  "isActive": true,
  "dob": "1964-30-08"
}
```

Document 2

```
{
  "id": "2",
  "fullName": "Sarah Jones",
  "isActive": false,
  "dob": "2002-02-18"
}
```

Document 3

```
{
  "id": "3",
  "fullName":
  {
    "first": "Adam",
    "last": "Stark"
  },
  "isActive": true,
  "dob": "2015-04-19"
}
```

Images extracted from: <https://dzone.com/articles/a-primer-on-open-source-nosql-databases>

<https://lennilobel.wordpress.com/2015/06/01/relational-databases-vs-nosql-document-databases/>

Column Databases

Row Storage

Last Name	First Name	E-mail	Phone #	Street Address

Columnar Storage

Last Name	First Name	E-mail	Phone #	Street Address

Graph Databases

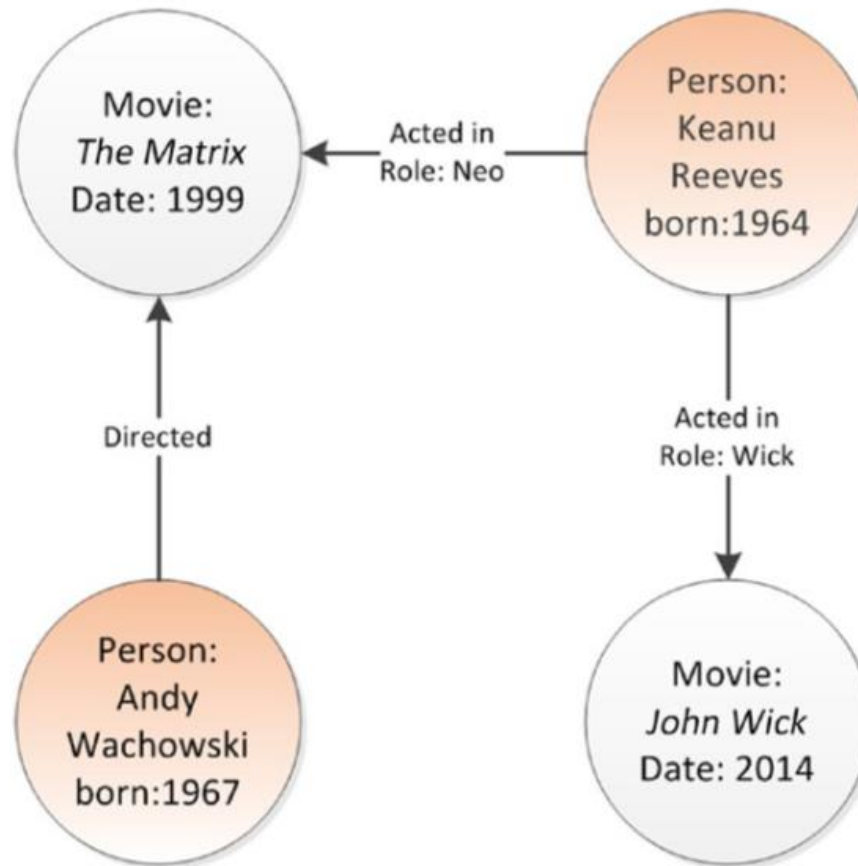


Image extracted from "Guy Harrison, Next Generation Databases, Apress, 2015"

Recaps Software Engineering

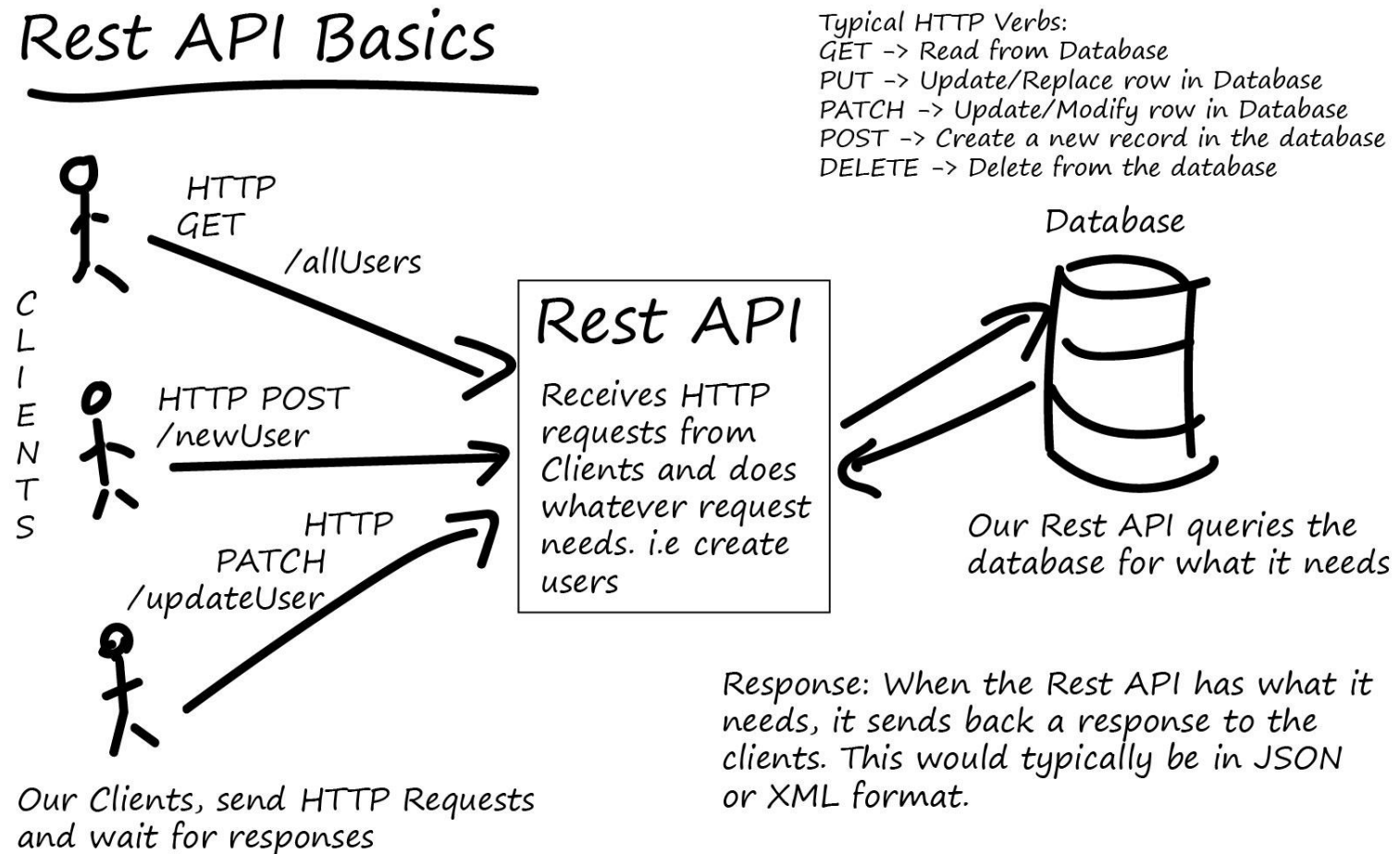
- Functional and non-functional requirements
- Use cases definitions
- UML Diagrams
- Some exercises on designing a complete application

Recaps on Java

- Hello World! In Java using an IDE (Eclipse/IntelliJ)
- Some Java Programming Exercises
- Connection to MySQL server: JDBC (using Maven for handling dependencies)
- Simple application connecting a relational DB which export its functionalities using RESTful API.

Recaps on Java

Rest API Basics



Modern NoSQL Infrastructures

Type	Example	
Key-Value Store	 redis	 riak
Wide Column Store	 H-BASE	 cassandra
Document Store	 mongoDB	 CouchDB <i>relax</i>
Graph Store	 Neo4j	 InfiniteGraph The Distributed Graph Database

amazon
DynamoDB



Learning Outcomes: *Knowledge*

At the end of the course:

- The student will have acquired knowledge about ***methodologies*** and ***tools*** and for the design of ***non-relational databases***.
- The student will acquire knowledge about the ***architectures***, ***performances*** and ***costs*** of modern infrastructures for the management of complex data.
- The student will be able to correctly ***set up*** a ***project*** for the management of multi-structured and large data, integrating it into a ***real computer application*** and choosing in an appropriate manner the design and implementation strategies.

Assessment Criteria of Knowledge

Group activities will be proposed to assess theoretical and practical knowledge.

Group activities will be proposed to the **working groups** with the objective of:

- **deepening** of theoretical and technical issues
- **implementing** of technical projects

Periodic classroom discussions between the teacher and the group of students developing the above activities will be organized.

Skills

At the end of the course the student will be able to:

- ***Design*** a non-relational database based on the ***requirements*** (functional and non-functional) of a specific ***application***.
- Use modern ***technological infrastructures*** for the management of non-relational databases (Redis, MongoDB, Neo4j)

Assessment Criteria of Skills

During lab class:

- The student will be shown how to ***install*** and ***configure*** some of the modern technological infrastructures for the management of non-relational databases.
- ***Practical activities*** will be proposed for the creation, management and querying of different non-relational databases.
- Group activities will be proposed for the ***in-depth study*** of technical issues and for the implementation of educational projects.

Prerequisites

- Programming in **JAVA** (including the use of an IDE and RESTful API)-> Skills acquired in ***Programmazione Avanzata***
- Design and query of ***relational databases***
- Basics of ***Software Engineering*** (including realization of UML Diagrams)
- Basics of ***Unix-Based*** Operating Systems

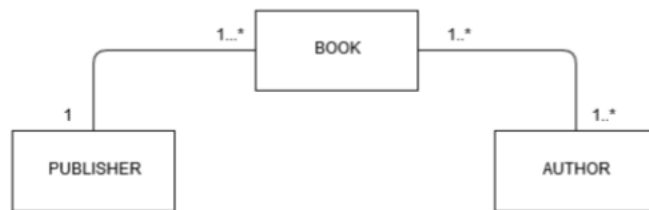
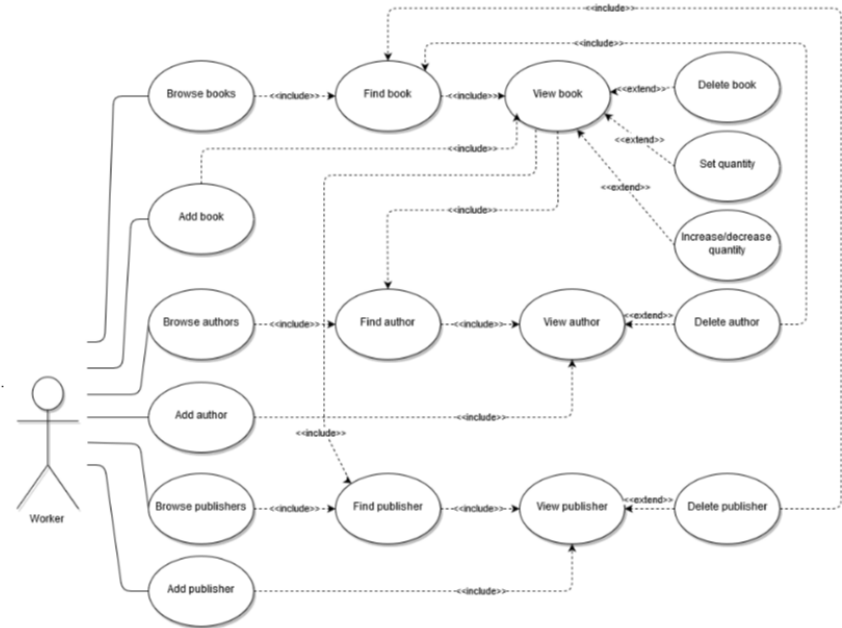
About Software Engineering Skills

1.1 Functional requirements

- The system has to allow the user to add and delete a book and modify its quantity;
- The system has to allow the user to add and delete a publisher;
- The system has to allow the user to add and delete an author.

1.2 Non functional requirements

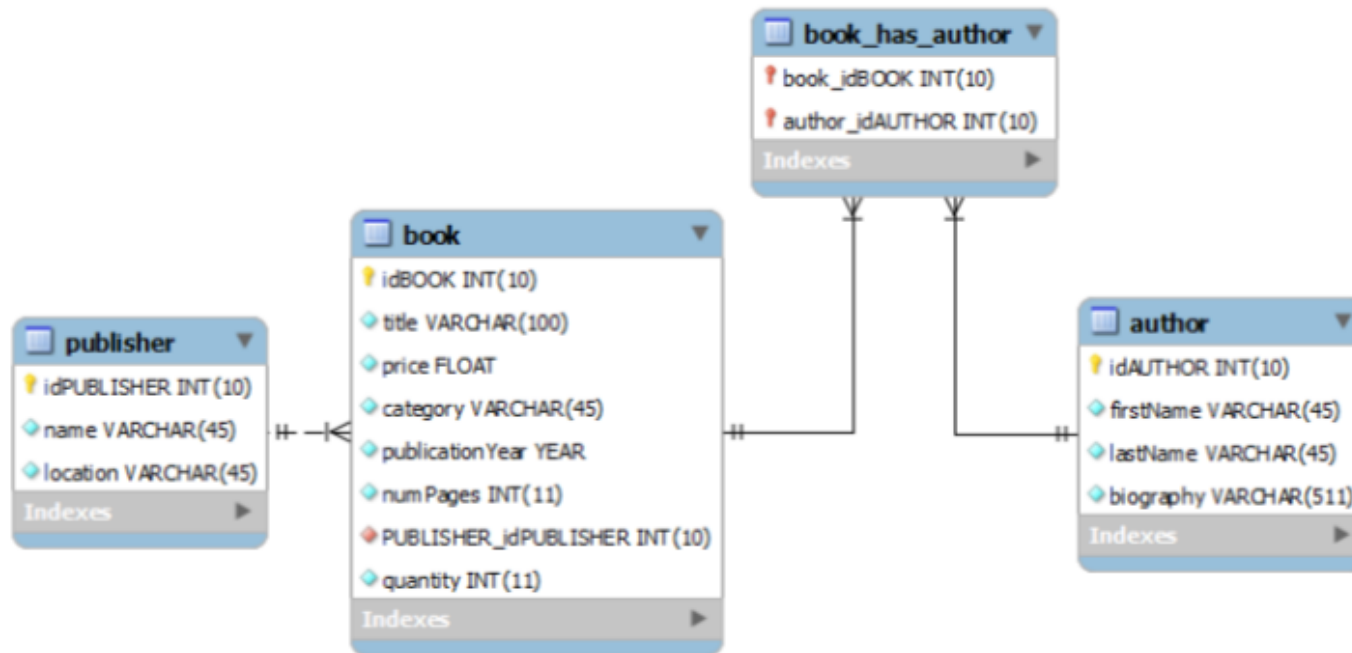
- Performance: The software has to be fast to avoid delays when a customer asks for a book;
- Integrity: The data integrity is crucial to avoid to give wrong information to the customers;
- Usability: The application must be user friendly and intuitive to be easily used by the workers.



UML Use Case Diagrams

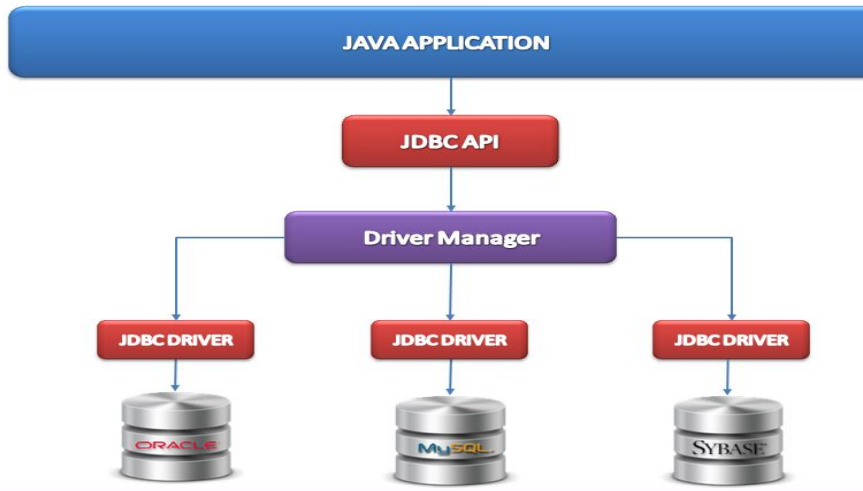
UML Analysis Class Diagrams

About Relational Databases

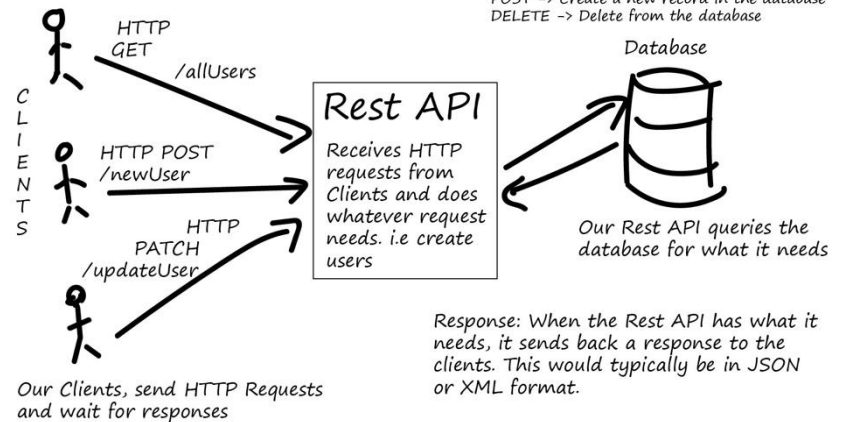


E-R Diagram

About (JAVA) Skills



Rest API Basics



Teaching Method

The course will be held in *face to face!*

The teacher will provide in advance (hopefully) the slides used during the lessons (with suggestions to the book chapter to be read).

The course will be held entirely in *English*.

Tutoring hours (two hours per week) will be provided each *Monday (REMOTE MODE)* (17:00-19:00) *by request*.

Book your meeting (MANDATORY) with teacher here:

<https://calendar.app.google/6whAiM6wNSZuxqW58>

The teacher will be available after the class for a *Q&A session*.

The E-learning Platform

We will exploit the Google GSuite service for the activities related to the course (materials, tests, projects).

Each student can login to the service with his/her own UNIFI credentials (check details here

<https://it.unipi.it/en/risorse/software/google-workspace-2/>

Once logged in, select the Classroom Service:



Duo



Google Keep



Jamboard



Classroom



Earth



Raccolte

From the + button Join a class (specify Class Code **7akhtmnu**) or
Use this link <https://classroom.google.com/c/ODg2MDc5MTY0OTY2?cjc=7akhtmnu>

Studying Materials

- Slides provided by the Teacher (mostly self-contained)
- Scientific articles provided by the teacher
- Official Documentation of the NoSQL DBMSs.
- Recommended Books:
 - “Guy Harrison, Next Generation Databases, Apress, 2015”
 - “Dan Sullivan, NoSQL For Mere Mortals, Addison-Wesley, 2015”
 - “Andreas Meier, Michael Kaufmann , SQL & NoSQL databases : models, languages, consistency options and architectures for big data management, 2019”
 - “Felipe Cardeneti Mendes, Piotr Sarna, Pavel Emelyanov, Cynthia Dunlop, Database Performance at Scale, Apress Open, 2023”

Check available books at: <https://onesearch.unipi.it>

THE HISTORY OF THIS COURSE...

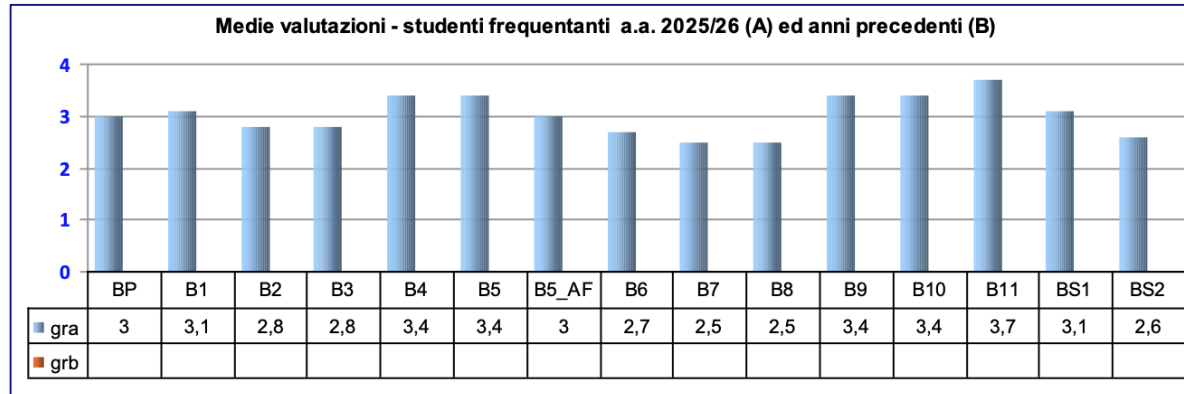
2023-2024 A.Y. Experience: Exam Results

Year	Average Grade	Number of Students
2026	26.49	100
2025	26.97	78
2024	25.26	69
2023	25.98	65
2022	26.99	93
2021	27.82	57
2020	28.58	83

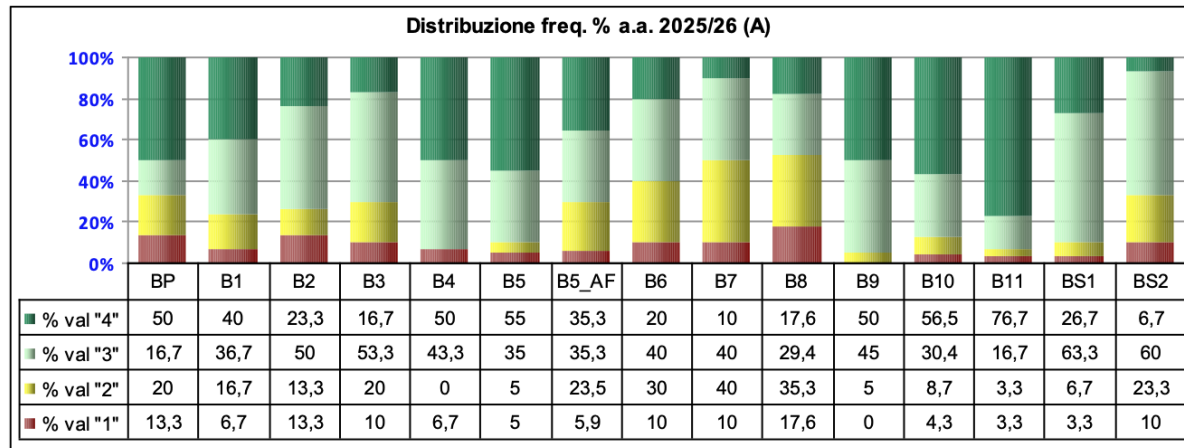
- The **percentage** of project discussed during the first exam session was equal to around 62% (35 groups in total, 22 discussions)
- The **percentage** of students attending the course (85) who **passed the exam** after the first exam session (62) was equal to **70 % (50% in the previous year)**.

2025-2026 A.Y. Experience: Students' Evaluation (CE)

Graf.1

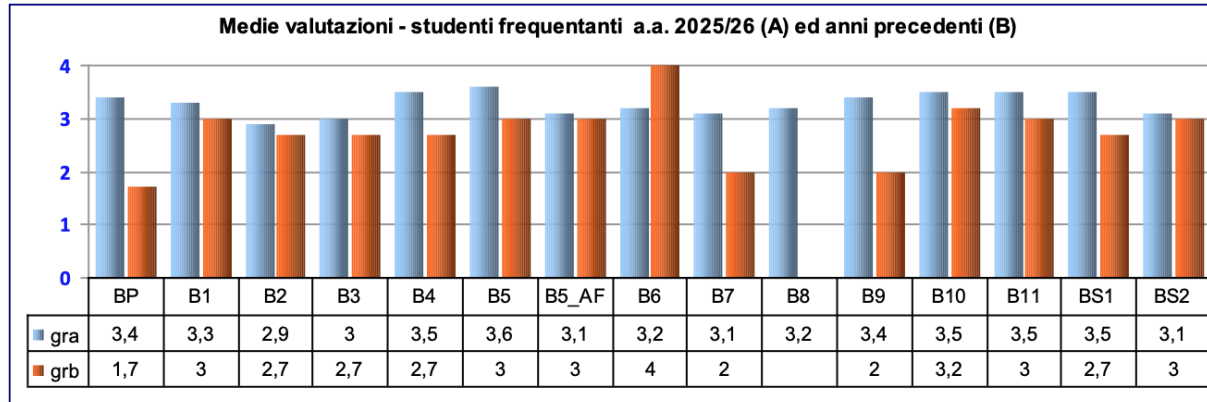


Graf.2

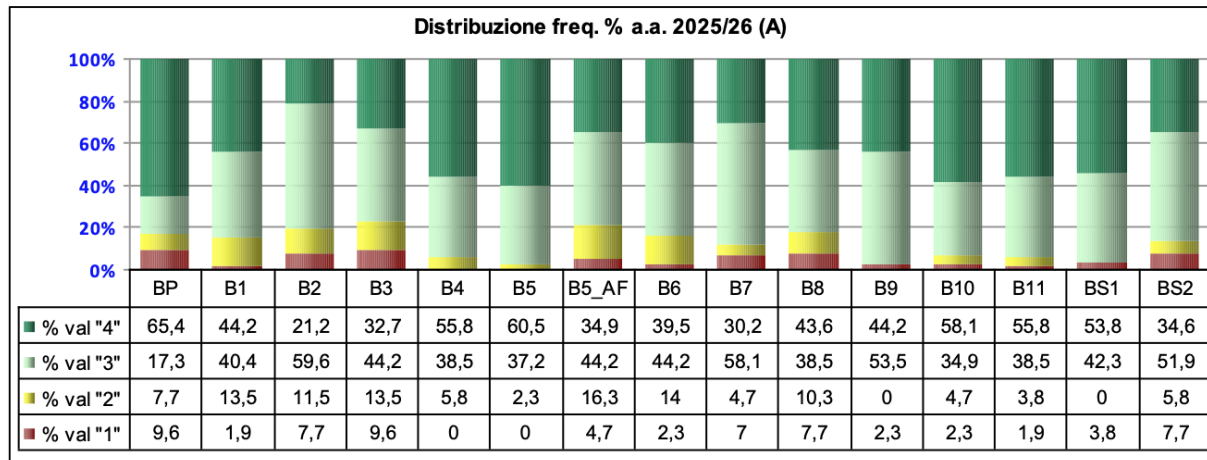


2026-2026 A.Y. Experience: Students' Evaluation (AIDE)

Graf.1



Graf.2



2025-2026 Course Implementation

From Now till the end of October/ Beginning of November 2026:

- Classes for introducing the main NoSQL architecture and strategies.
- Recaps of Software Engineering and Java (with exercises)
- Examples of Applications based on NoSQL DB architectures
- Introduction to the Project and Student Group Definition (1 round)

2026-2027 Course Implementation

During November 2026:

- Introduction to the main features of NoSQL Infrastructures (Redis, MongoDB, Neo4j)
- Student Group Definition (2 round)

By the end of November 2026 up to the end of academic year:

- Student Group Definition (3 round)
- Project idea discussion
- Project development

Possible Extra Classes

The week of Dec. 14, 2026 to Dec. 18, 2026 is reserved for any extra class hours as needed (i.e. less than 90 hours delivered).

The Enrolment form

Please fill and submit the following form:

<https://forms.gle/ntr4UDeNwxvMdVmJ8>

It's mandatory for attending labs and for the cooperative group activities

About the exam

- Discussion of the project (50%)
- Written test on the theoretical parts of the course (50%):
three open questions, 40 minutes.

About the Project

Design and develop of an Application interacting with NoSQL Databases

- Start with an idea and a draft of application requirements, use case diagrams and data entities (quick discussion in the classroom during the course or during meeting slots, ***to be approved by the teacher***)
- Refine requirements and use cases
- Define data entities and relationships by means of UML analysis class diagrams
- Provide the design of the Data Base (at least two of the main non relational models must be considered when defining the requirements. ***Document DB model is mandatory***)
- Define the main queries on the Data Base(s)
- Implement and deploy the application, hopefully on the Virtual Lab
- Test the application (by running the implemented RESTful APIs, details will be provided).
- Write a complete documentation resuming the above items and including an APIs documentation.

Some Advices for the Project

- ***Attend carefully*** the lessons during the first 6-8 weeks, especially pay attention to the examples of application discussed by the teacher (often the best projects of past students).
- ***Deepen*** your ***skills*** in Java programming and Software Engineering and make the suggested exercises.
- ***Ask support*** to the teacher whenever required.
- Come to the meeting with the teacher for asking for additional clarifications, explanations, advices, and resolving doubts.

Rules for the Project

- The project must be developed ***in groups of 3 students*** (for special cases, talk with the teacher)
- ***No reviews*** are allowed, students will receive a number of examples of past projects.
- The project must be ***discussed before the*** written test (the date will be fixed by the teacher)
- The ***final documentation*** must be submitted to the teacher ***in advance*** (deadline will be fixed some days before the discussion)
- Avoid to ***involve*** the teacher for ***solving problems*** among group members.

Rules for Project Evaluation

Overall Project Evaluation:

- 25 % for the Idea, requirements definitions, the entity-relationship model (and UML diagrams)
- 40% for DB design and query definition
- 25% for the implementation of APIs
- 10% for the clarity of the overall documentation

The ***individual assessment*** will depend on the overall evaluation of the project and the answers given by the specific student during the ***project discussions***.

The Self-Assessment Survey

Each student will be required to fill a survey for *self-assessing* their technical and theoretical skills.

<https://forms.gle/TgH9CWhwfdcFhMDd8>

The results of the survey will be used for *better focusing the teaching activities*.

The data collected will be used only for *statistical, teaching* and *research* purposes.

Publications (if any) will only report analysis that will use aggregated and anonymized data .

The *data will not be transferred* to third parties or to user profiling companies that may use them for commercial purposes.

If personal data (such as Name, Surname and e-mail address) will be provided, the owner may at *any time request to delete the data* and not to use them for further analysis.

Contacts

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Web: <https://sites.google.com/site/ducangepietro>